



200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

RECEIVED

JUL 07 2014

7/2/2014

Mr. Dave Schipper
Department of Environmental Quality
350 Ottawa N.W.
Grand Rapids, MI 49503

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Subject: Amendment to Existing RMP

Dear Mr. Schipper:

The City of Cadillac would like to add a method to Section III (Land Application Plan) Part B-4 of our RMP whereas we can surface apply biosolids. We would like to have this option, if the sub-surface injection cannot be carried out. We meet the criteria for "Exceptional Quality" Biosolid as listed in Part 24;

- (i) Pollutant ceiling concentrations in R 323.2409(5)(a)
- (ii) Pollutant concentrations in R 323.2409(5)(c)
- (iii) Vector attraction reduction options in R 323.2415(4) (a); *we had 54.11% volatile reduction in 2013 with primary digester heated to average of 96.20 °F using external heat exchanger and retention time of 22.2 days. And class A pathogen reduction alternatives in R 323.2414(2)(a); the density of salmonella sp. bacteria in the biosolids was <3 MPN per 4 grams of total solids (dry weight basis) for 2013.*

The following analytical data represents the average contents of biosolids in our most recent analysis. Please find the attached documents for the complete list of the variables analyzed.

Recent biosolids analysis in mg/kg (dry weight basis)

Name	Concentration (mg/kg) Dry Weight	Pollutant Concentrations (Table 3, 40 CFR 503.13) (monthly average)	Ceiling Concentrations* (Table 1, 40 CFR 503.13) (daily maximum)
Arsenic	15.3	41 mg/kg	75 mg/kg
Cadmium	2.7	39 mg/kg	85 mg/kg
Copper	706	1500 mg/kg	4300 mg/kg
Lead	<50	300 mg/kg	840 mg/kg
Mercury	0.72	17 mg/kg	57 mg/kg
Molybdenum	23	N/A**	75 mg/kg
Nickel	18.6	420 mg/kg	420 mg/kg
Selenium	4.2	100 mg/kg	100 mg/kg
Zinc	1060	2800 mg/kg	7500 mg/kg
Nitrogen Concentration		N/A	N/A

* Biosolids may not be land applied if any pollutant exceeds these values.

** EPA has temporarily removed molybdenum limits from Table 2, Table 3 and Table 4

RMP
MOD

If you have any questions, please do not hesitate to contact me.

Sincerely,



Hamid Rahbarnoohi, Water Resources Supervisor

City of Cadillac

200 N. Lake Street

Email: hrahbarnoohi@cadillac-mi.net; Phone: 231-775-2841; Fax: 231-775-2876

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JUL 07 2014

**DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS**



Testing • Research • Consulting

Multiple Analysis Report

Sample: 268372

July 24, 2013

Ms. Amy Vail
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Digester #4
1121 Plett Rd.
Cadillac, MI 49601

Dear Ms. Vail:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 10:26 AM, 04/16/2013, and collected at 2:00 PM, 04/15/2013:

<u>TEST - METHOD</u>	<u>RESULT</u>	<u>MDL*</u>	<u>Date Complete</u>
Fecal Coliform-SM9221E (MPN)	1,160 mpn/g dry	1 MPN/g dry	04/19/2013
Enteric Virus-EPA 625/R-92/013	<1 pfu/4g dry	1 pfu/4g dry	07/23/2013
Solids, total-SM 2540G	2.483%	0.005%	05/13/2013
Salmonella-EPA 1682	<3 MPN/4 g dry	3 MPN/4 g dry	04/18/2013
Helminth Ova-EPA/625/R-92/013	<1 ovum/4g dry	1 ovum/4 g dry	07/19/2013

*Minimum Detection Level

This testing was completed by J.B. and N.R. Please feel free to contact us if we can be of further service to you.

Sincerely,

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp

RECEIVED

JUL 07 2014

**DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS**

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

RECEIVED KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

JUL 07 2014

Project

Description: **Analysis of one Spring Biosolids sample.**

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	04/14/14	BPK	
Prep, metals	Completed		EPA 200.7/8	04/03/14	ZZW	
Aluminum, total	2840	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Arsenic, total	15.3	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 1
Cadmium, total	2.7	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 0.2
Calcium, total	18,900	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Chromium, total	81.8	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 4.6
Copper, total	706	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Lead, total, by ICP	<50	mg/kg total solids	EPA 200.8	04/15/14	ZZW	
Magnesium, total	3040	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Mercury, total	0.72	mg/kg total solids	EPA 7471A	04/14/14	BPK	Detection limit = 0.05
Molybdenum, total	23.0	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Nickel, total	18.6	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Potassium, total	90.8	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Selenium, total	4.2	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 1
Silver, total	<4.6	mg/kg total solids	EPA 6010B	04/14/14	JHB	
Sodium, total	4630	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Zinc, total	1060	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
TCLP extraction	Completed		EPA 1311	04/02/14	JWW	
TCLP report	See comment			04/15/14	SRM	This material does not exhibit the Toxicity Characteristic with respect to the requested TC parameters.
ZHE extraction	Completed		EPA 1311	04/02/14	JWW	
Chloride	6780	mg/kg total solids	EPA 300.0A	04/04/14	ALK	Detection limit = 190
Nitrogen, ammonia	36,100	mg/kg total solids	EPA 350.1	04/02/14	JWW	Detection limit = 1500
Nitrogen, nitrate	<37	mg/kg total solids	EPA 353.2	04/03/14	ALK	
Nitrogen, total kjeldahl	81,100	mg/kg total solids	EPA 351.2	04/07/14	ALK	Detection limit = 2600
Phosphorus, total (as P)	41,800	mg/kg total solids	SM 4500-P B5,E	04/08/14	JHB	Detection limit = 3000
Sulfate	2110	mg/kg total solids	EPA 300.0A	04/04/14	ALK	Detection limit = 370
Waste pH	7.7	S.U.	EPA 9045C	04/09/14	AMG	
Solids, total	2.74	% by weight	EPA 160.3	04/03/14	AMG	Detection limit = 0.01
Solids, volatile	57.70	% of total solids	EPA 160.4	04/03/14	AMG	Detection limit = 0.01
Prior. Poll. volatiles	See below		EPA 8260	04/02/14	JAR	
Prep, VOA	Completed		EPA 5030	04/02/14	JAR	
1,1,1-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1,2,2-Tetrachloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1,2-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichloropropane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,3-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,4-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2-Chloroethylvinyl ether	<20,000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Acrolein	<10,000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Acrylonitrile	<4000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Benzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromodichloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromoform	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromomethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Carbon tetrachloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloroform	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Cis-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Dibromochloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Ethylbenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Methylene chloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Tetrachloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Toluene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trans-1,2-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trans-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trichlorofluoromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Vinyl chloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Prior. Poll. acids	See below		EPA 8270	04/09/14	GMB	
Prior. Poll. base-neutrals	See below		EPA 8270	04/09/14	GMB	
Prep, SV Acid/BN	Completed		EPA 3545	04/08/14	GMB	
1,2,4-Trichlorobenzene 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,2-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,2-Diphenylhydrazine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,3-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,4-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,3,7,8-TCDD by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4,6-Trichlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dichlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dimethylphenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dinitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dinitrotoluene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,6-Dinitrotoluene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Chloronaphthalene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Chlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Methyl-4,6-dinitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Nitrophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
3,3'-Dichlorobenzidine	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4-Bromophenyl phenyl ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Chloro-3-methylphenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Chlorophenyl phenyl ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Nitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Acenaphthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Acenaphthylene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzidine	<70,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(a)anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(a)pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(b)fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(ghi)perylene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(k)fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroethoxy)methane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroethyl)ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroisopropyl)ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-ethylhexyl)phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Butylbenzyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Chrysene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Di-n-butyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Di-n-octyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dibenzo(ah)anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Diethyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dimethyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Fluorene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorobenzene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorobutadiene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorocyclopentadiene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachloroethane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Indeno(123cd)pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Isophorone	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodi-n-propylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodimethylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodiphenylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Naphthalene by Method 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Nitrobenzene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Pentachlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Phenanthrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Phenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Prior. Poll. pest./PCB's	See below		EPA 8270	04/09/14	GMB	
4,4'-DDD	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4,4'-DDE	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4,4'-DDT	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Aldrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Alpha-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Beta-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Chlordane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Delta-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dieldrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan I	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan II	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan sulfate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endrin aldehyde	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Gamma-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Heptachlor	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Heptachlor epoxide	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1016 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1221 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1232 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1242 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1248 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1254 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1260 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Toxaphene	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **402028-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	Completed		EPA 200.7	04/10/14	BPK	
Prep, Hg	Completed		EPA 7470A	04/03/14	BPK	
Prep, metals	Completed		EPA 200.7/8	04/03/14	ZZW	
Arsenic, total, by ICP	<0.5	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.
Barium, total	1.37	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 100 mg/L.
Cadmium, total	<0.025	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 1.0 mg/L.
Chromium, total	<0.05	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.
Copper, total	<0.1	mg/L	EPA 200.7	04/10/14	BPK	
Lead, total, by ICP	<0.25	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **402028-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Mercury, total	<0.05	mg/L	EPA 7470A	04/03/14	BPK	TC regulatory limit is 0.2 mg/L.
Selenium, total, by ICP	<0.5	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 1.0 mg/L.
Silver, total	<0.025	mg/L	EPA 200.7	04/04/14	JHB	TC regulatory limit is 5.0 mg/L.
Zinc, total	1.29	mg/L	EPA 200.7	04/10/14	BPK	
TC Semi-Volatiles	See below		EPA 8270	04/03/14	GMB	
Prep, SV Acid/BN	Completed			04/03/14	GMB	
Prep, herbicides	Completed		EPA 8151	04/07/14	GMB	
2,4,5-Trichlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 400 mg/L.
2,4,6-Trichlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 2.0 mg/L.
2,4-Dinitrotoluene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.13 mg/L.
Cresols	<0.2	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 200 mg/L.
Hexachlorobenzene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.13 mg/L.
Hexachlorobutadiene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.5 mg/L.
Hexachloroethane	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 3.0 mg/L.
Nitrobenzene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 2.0 mg/L.
Pentachlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 100 mg/L.
Pyridine	<0.2	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 5.0 mg/L.
TC Pesticides by ECD	See below		EPA 8081	04/04/14	GMB	
Prep, ECD	Completed		EPA 3510	04/03/14	GMB	
Chlordane	<0.02	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.03 mg/L.
Endrin	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.02 mg/L.
Heptachlor	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.008 mg/L.
Heptachlor epoxide	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.008 mg/L.
Lindane by ECD	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.4 mg/L.
Methoxychlor	<0.02	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 10.0 mg/L.
Toxaphene	<0.04	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.5 mg/L.
TC Herbicides	See below		EPA 8151	04/08/14	GMB	
2,4,5-TP (Silvex)	<0.2	mg/L	EPA 8151	04/08/14	GMB	TC regulatory limit is 1.0 mg/L.
2,4-D	<0.2	mg/L	EPA 8151	04/08/14	GMB	TC regulatory limit is 10.0 mg/L.

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **ZHE**

Sample Time :

KAR Sample No. : **402028-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Volatiles	See below		EPA 8260	04/03/14	JAR	
Prep, VOA	Completed		EPA 5030	04/03/14	JAR	
1,1-Dichloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.7 mg/L.
1,2-Dichloroethane	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **ZHE**

Sample Time :

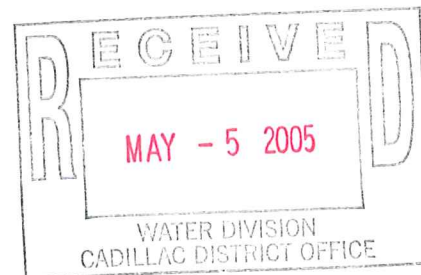
KAR Sample No. : **402028-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 7.5 mg/L.
Benzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Carbon tetrachloride	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Chlorobenzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 100 mg/L.
Chloroform	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 6.0 mg/L.
Methyl ethyl ketone	<2	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 200 mg/L.
Tetrachloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.7 mg/L.
Trichloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Vinyl chloride	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.2 mg/L.

CITY OF
CADILLAC
MICHIGAN

200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

RMP Dev. file
SOC ✓



May 5 , 2005

Mr. Mike Person
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac , Michigan 49601

SUBJECT : RMP Certification

Dear Mr. Person,

The City of Cadillac would like to certify that the current residual management practices are in accordance with the approved RMP. We have added some fields to our RMP and we will be sending you an updated list.

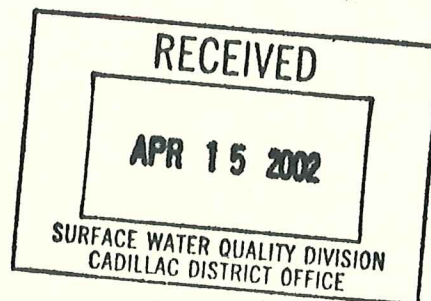
If you have any questions please feel free to call me.

Jeffrey M. Dietlin
Water Resources Supervisor
Phone # : (231) 775 - 2841
Fax # (231) 775 - 2876



April 15 , 2002

Mr. Mike Person
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac , Michigan 49601



SUBJECT : RMP Application

Dear Mr. Person,

Enclosed please find the City of Cadillacs updated RMP. Enclosed are all of the changes that we discussed in our October 2001 meeting. As we discussed I am only sending you the report and not the information in the Appendixes because you already have this information .

The City recently reapplied for our NPDES permit and a lot of the information in the RMP was used for this.

I am personally sorry for the timeframe in which this was submitted.

If you have any questions please feel free to call me.

A handwritten signature in black ink, appearing to read "Jeffrey M. Dietlin".

Jeffrey M. Dietlin
Water Resources Supervisor
Phone # : (616) 775 - 2841
Fax # 231-775 - 2876



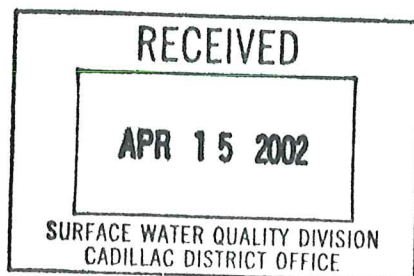
NPDES Permit Number or COC Number
MI0020257

State of Michigan Biosolids Land Application Program

Facility Name
CITY OF CADILLAC WASTE WATER TREATMENT PLANT

Residuals Management Program Development Document

General Instructions	Page	i
Section I - General Information	Pages	1 - 2
Section II - Biosolids Characterization	Pages	3 - 4
Section III - Land Application Plan	Pages	5 - 16
Signature Page	Page	17
Definitions	Pages	18 - 20
Additional Information	Page	21
Appendix	Pages	22 - 41



SURFACE WATER QUALITY DIVISION
MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

Revised
RMP

PURPOSE

The Michigan Biosolids Program establishes standards for the land application and beneficial recycling of biosolids in the state. Any treatment works treating domestic sewage proposing to land apply biosolids in the state shall prepare the enclosed Residuals Management Program (RMP) Guidance Document to obtain authorization. Authorization is required under the auspices of either an individual NPDES Permit or General Permit "Certificate of Coverage" (COC), dependent upon the type of permit held by the generating facility proposing to land apply.

For assistance on completing any section in this document you may contact the appropriate DEQ Surface Water Quality Division (SWQD) Office listed in Appendix A.

Industrial or commercial facilities with process wastewater entering the waste stream do not qualify for issuance of a biosolids land application permit. Contact the DEQ Waste Management Division in the appropriate district office for information on handling commercial or industrial wastes.

AUTHORITY

The Part 24 Rules, Land Application of Biosolids, of Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended, (NREPA) provide authority to issue Biosolids Land Application Permits. Completion and subsequent approval of this document meets the RMP requirements contained in Rule 2403, Land Application Permit of the Part 24 Rules. Failure to prepare and submit an RMP prior to land applying biosolids is a violation of Part 24 Rules and may result in penalties for such violation.

Rule 2403.

- (1) A generator or distributor shall have a valid permit before commencing any biosolids land application or distribution activity in the state of Michigan.
- (2) A generator or distributor shall submit a permit application on a form provided by the DEQ not less than 180 days before expiration of an existing permit, commencement of biosolids land application, or distribution of a biosolids derivative, except at retail.
- (3) All of the following provisions apply to a residuals management program:
 - (a) A generator shall submit a residuals management program for approval by the DEQ as required by its permit.
 - (b) A residuals management program submitted to the DEQ shall include all of the following information:
 - (i) Size and type of generating facility.
 - (ii) One year of records representing the volume and concentrations of pollutants in the biosolids.
 - (iii) Treatment process origin, for example, primary or secondary treatment and the volume of biosolids generated from each process.
 - (iv) A description of the treatment processes.
 - (v) Storage volume.
 - (vi) Transportation methods and spill prevention plan.
 - (vii) Land application method.
 - (viii) Land application site list.
 - (ix) Land application plan.
 - (x) Pathogen reduction method.
 - (xi) Vector attraction reduction method.
 - (xii) Monitoring program.
 - (c) Upon approval by the DEQ, the generating facility shall implement the approved residuals management program.
 - (d) A generating facility may modify the approved residuals management program by submitting a proposed modification to the DEQ for approval. The modification shall become effective upon approval by the DEQ.
- (4) A person shall land apply biosolids or prepare biosolids for land application in accordance with the requirements established in these rules.

The Michigan Department of Environmental Quality (DEQ) will not discriminate against any individual or group on the basis of race, sex, religion, age, national origin, color, marital status, disability, or political beliefs. Questions or concerns should be directed to the Office of Personnel Services, PO Box 30473, Lansing MI 48909

RESIDUALS MANAGEMENT PROGRAM**SECTION I – GENERAL INFORMATION**

PLEASE TYPE OR PRINT

1. NPDES or COC NUMBER MI0020257			4. FACILITY MAILING ADDRESS Street Address or P.O. Box (or check box to use address corresponding to item number ☒ 2 ☐ 3)																				
2. GENERATOR NAME AND MAILING ADDRESS Name <u>City of Cadillac</u> Superintendent, CEO, Principal Executive Officer Name <u>JEFFREY M. DIETLIN</u> Street Address or P.O. Box 200 NORTH LAKE STREET			Additional Street Address or P.O. Box Information																				
City or Village <u>CADILLAC</u>	State <u>MI</u>	ZIP Code <u>49601</u>	City or Village State ZIP Code																				
Telephone (include area code) <u>231 -775 -2841</u>			5. BIOSOLIDS CONTACT PERSON Name (if different than Item 2) Title																				
3. FACILITY NAME AND LOCATION (if different) Name <u>City of Cadillac</u> Street Address 1121 PLETT ROAD Additional Street Address			ADDRESS Street Address or P.O. Box (or check box to use address corresponding to item number ☒ 2 ☐ 3 ☐ 4)																				
City or Village <u>CADILLAC</u>	State <u>MI</u>	ZIP Code <u>49601</u>	City or Village State ZIP Code																				
Latitude (nearest 15 seconds) <u>N44 15.915</u>			Longitude <u>W085 23.851</u>																				
Telephone (include area code) <u>231-775-2841</u>			Telephone (include area code)																				
Fax Number (include area code) <u>231-775-2876</u>			Fax Number (with area code)																				
Email Address (if any) <u>WWTPR@CADILLAC-MI.NET</u>			6. Type of Treatment Facility Publicly owned treatment works ☒ Privately owned treatment works ☐ Federally owned treatment works ☐ Blending or treatment operation ☐ State owned treatment works ☐ Tribal owned treatment works ☐ Other ☐ explain:																				
7. LAND APPLICATION REPORT FORMS Check the box that corresponds to the address (above) to which forms should be mailed. ☒ 2 – Applicant Name & Mailing Address ☐ 3 - Facility Name & Location ☐ 4 - Facility Mailing Address ☐ 5 - Contact Person SEND FORMS TO THE ATTENTION OF: <u>Jeffrey M. Dietlin</u>																							
8. Provide the names, addresses and telephone numbers of the Land Application Contractors used by the facility, if applicable. <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 20%;">Name of Contractor:</td> <td><u>SYNAGRO MIDWEST</u></td> </tr> <tr> <td>Address:</td> <td><u>323 MARTINDALE</u></td> </tr> <tr> <td>City, State, ZipCode:</td> <td><u>SPARTA MI ,49345</u></td> </tr> <tr> <td>Telephone:</td> <td><u>616-887-1144</u></td> </tr> <tr><td colspan="2"> </td></tr> <tr> <td>Name of Contractor:</td> <td></td> </tr> <tr> <td>Address:</td> <td></td> </tr> <tr> <td>City, State, ZipCode:</td> <td></td> </tr> <tr> <td>Telephone:</td> <td></td> </tr> </table>						Name of Contractor:	<u>SYNAGRO MIDWEST</u>	Address:	<u>323 MARTINDALE</u>	City, State, ZipCode:	<u>SPARTA MI ,49345</u>	Telephone:	<u>616-887-1144</u>			Name of Contractor:		Address:		City, State, ZipCode:		Telephone:	
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City, State, ZipCode:	<u>SPARTA MI ,49345</u>																						
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Name of Contractor:																							
Address:																							
City, State, ZipCode:																							
Telephone:																							

RESIDUALS MANAGEMENT PROGRAM**SECTION I – GENERAL INFORMATION**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
9. RESIDUALS HANDLING Provide total English dry tons per 365-day period of biosolids handled under the following practices: Amount generated at the facility: <u>327</u> Amount received from off site: <u>0</u> Amount treated on site (including blending): <u>327</u> Amount sold or given away in a bag or other container for application to the land: <u>0</u> Amount of bulk biosolids shipped off site for treatment or for sale/give-away in a bag or other container for application to the land: <u>0</u> Amount applied to land in bulk form: <u>327</u> Amount fired in biosolids incinerator: <u>0</u> Amount sent to municipal solid waste landfill: <u>0</u> Amount used or disposed by another practice: <u>0</u> Describe: _____ _____ _____ _____ _____ _____	10. RESIDUALS STORAGE Enter the volume of residual storage capacity at this facility. 1.6 ☒ million gallons or ☐ cubic feet 11. WASTEWATER FLOW RATE Provide design and actual flow rates in million gallons per day (MGD). Design flow <u>3.2</u> MGD Average Actual Flow <u>2.3</u> MGD 12. SERVICE AREA Enter the population serviced or NA if not applicable. <u>11500</u> 13. INDUSTRIAL PRETREATMENT PROGRAM Does the generating facility have, or are they required to have, an Industrial Pretreatment Program? (See Appendix A) ☒ Yes - On attached sheet provide list of Significant Users and list of pollutants monitored. No - Continue with item 14. 14. SEPTAGE Does your facility accept septage? Yes - Estimate amount received monthly ☒ No - Continue with item 15. 15. BLENDING AND TREATMENT FACILITIES Are (or will) biosolids from another facility (not including septage) be sent to your facility for treatment (including blending) or placement in a bag or other container for sale or give-away? Yes - Contact the Pretreatment and Biosolids Unit, Lansing ☒ No - Continue with item 16.
16. SOLIDS FLOW DIAGRAM, NARRATIVE DESCRIPTION, and PRODUCTION DATA Provide a flow diagram (using 8 1/2" x 11" paper if possible) showing the solids handling process through the facility. Identify all treatment units and sampling points. Please attach this diagram. Include a narrative description of all solids handling through the facility and provide solids production data of solids unit processes. Provide detailed information on residuals stabilization process including all solids handling unit processes (i.e. collection, dewatering, digestion, chemical feeds, storage). Include a narrative that briefly describes the history of the residuals handling practices at the treatment facility and any future plans for upgrade. OPTIONAL- To assist in calculating production data for your facility, Excel spreadsheets of various processes have been developed by the DEQ Operator Training Unit and are available from the appropriate DEQ SWQD Office listed in Appendix A or via the biosolids homepage at http://www.deq.state.mi.us/swq/ . If you are an industrial or commercial facility treating domestic sewage sludge only, the line diagram shall include all operations contributing wastewater including process and production areas, sanitary flows, cooling water and storm water runoff. Include a narrative, which provides a brief description of the manufacturing processes.	

RESIDUALS MANAGEMENT PROGRAM**SECTION II – BIOSOLIDS CHARACTERIZATION**

PLEASE TYPE OR PRINT

FACILITY NAME Caddillac Wast Water Treatment Plant				NPDES or COC PERMIT NUMBER MI0020257			
1. BIOSOLIDS CHARACTERISTICS Report one year biosolids monitoring data, and in no case less than three sampling events for the following Part 24 rules required parameters. Provide the actual analytical data sheets as an attachment. All sampling shall be representative of the biosolids being proposed to be applied to the land and in accordance with the minimum sample frequencies provided on the Page 10. Analytical methods shall be in accordance with R 323.2406 (2) Methods for Biosolids. Data that is available for parameters not specifically listed on this page shall be provided in the Pollutants of Concern table on page 4, Item 3 of this section.							
Parameter	Average Monthly Concentration	Maximum Monthly Concentration	Units	Number of Analyses	Method Detection Limit	Test Method	Sample Type
Inorganics							
Total Solids	4.97	7.8	%	3		SM2540B	☐ Grab ☒ Composite
Total Arsenic	20.4	25.2	Mg/kg	3		EPA 6020	☐ Grab ☒ Composite
Total Cadmium	8.18	9.7	Mg/kg	3		EPA 6020	☐ Grab ☒ Composite
Total Copper	900	981	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Lead	135.3	142	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Mercury	1	1.4	Mg/kg	3		EPa7471A	☐ Grab ☒ Composite
Total Molybdenum	60.67	75	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Nickel	48.3	57	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Selenium	2.77	6.5	Mg/kg	3		EPA 7740	☐ Grab ☒ Composite
Total Zinc	1933.33	2090	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Nutrients							
Total Kjeldahl Nitrogen	5.27	5.5	%	3		EPA 351.2	☐ Grab ☒ Composite
Ammonium Nitrogen	2.6	2.6	%	3		EPA 350.1	☐ Grab ☒ Composite
Total Phosphorus	4.83	6.3	%	3		SM4500-PE	☐ Grab ☒ Composite
Total Potassium	2233.33	2470	mg/kg	3		EPA 6010B	☐ Grab ☒ Composite

RESIDUALS MANAGEMENT PROGRAM**SECTION II – BIOSOLIDS CHARACTERIZATION**

PLEASE TYPE OR PRINT

FACILITY NAME

Cadillac Waste Water Treatment Plant

NPDES or COC PERMIT NUMBER

MI0020257

2. Pollutants of Concern

Do you believe or suspect that pollutants (other than parameters listed on the previous page) could be present in the biosolids to be land applied at concentrations that could possibly make them unsuitable for land application?

☐ Yes - Describe the circumstances below and provide representative analytical data for those pollutants of concern in Item 3.

☒ No

3. Report any biosolids monitoring data from the last permit cycle for parameters not specifically listed on the previous page. Include the actual analytical data sheets as an attachment. Upon submittal review, additional monitoring may be required if SWQD has reason(s) to suspect that the information provided (or not provided) does not adequately characterize the residuals proposed to be land applied. For assistance on completing this section or determining the necessity for completing this section, you may contact the appropriate DEQ SWQD office (see Appendix A).

Parameter	Average Concentration	Maximum Concentration	Units	Number of Analyses	Method Detection Limit	Test Method	Sample Type
ALUMINUM	7116.7	7690	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
CALCIUM	32633	35100	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
CHROMIUM	125	143	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
MAGNESIUM	3978	4170	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
SILVER	63.3	80.9	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
SODIUM	3937	4150	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
CHLORIDE	4377	4480	mg/kg	3		EPA300A	☐ Grab ☒ Composite
SULFATE	675	1310	mg/kg	3		EPA300A	☐ Grab ☒ Composite
BIS (2-ETHYLHEXYL) PHTHALATE	62000	64000	ug/kg	2		EPA8270	☐ Grab ☒ Composite
METHYLENE CHLORIDE	1400	2800	ug/kg	2		EPA8270	☐ Grab ☒ Composite
PRIORITY POLLUTANTS	BELOW DET.	BELOW DET.	ug/kg	2		EPA8270	☐ Grab ☒ Composite
							☐ Grab ☐ Composite
							☐ Grab ☐ Composite
							☐ Grab ☐ Composite

RESIDUALS MANAGEMENT PROGRAM

SECTION III – LAND APPLICATION PLAN

Instructions for completing Section III.

The information provided in this Land Application Plan section outlines the generating facility's RMP. In addition to requirements set forth in the Michigan Part 24 Biosolids Rules, the information provided (except items marked **Optional**) represent enforceable commitments to manage the land application program in the manner indicated. If changes in circumstances arise, opportunities do exist to request modification to approved RMPs during the biosolids permit cycle.

Although land application contractors are commonly utilized to handle many operational aspects of land application programs, each facility is required to take an active role in managing and overseeing certain administrative aspects of their land application program. Although the DEQ may hold contractors responsible for certain violations of Part 24 rules, the generating facility is responsible for ensuring compliance with state and federal statutes and the conditions found in the approved RMP.

- A. **Biosolids Treatment** - Identify the exact alternatives used at the facility to meet Pathogen Reduction and Vector Attraction Reduction requirements and include the appropriate documentation to demonstrate compliance with those conditions. If your facility utilizes more than one alternative to meet the requirements, indicate all that apply. Provide additional information on attached sheets.
- B. **Procedures** – General programmatic information outlining the facilities biosolids operating procedures.
- C. **Site Information** – Information specific to your land application sites and overall site management.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****A. Biosolids Treatment**

PLEASE TYPE OR PRINT

FACILITY NAME CADILLAC WAS	NPDES or COC PERMIT NUMBER MI0020257
1. PATHOGEN / VECTOR ATTRACTION REDUCTION Describe, on this form or another sheet of paper, any treatment processes used at your facility to reduce pathogen and vector attraction properties in biosolids: Anaerobic digestion with sludge holding tanks. Most of our pathogen and vector attraction is taken care of in our anaerobic digester which we run at 96 degree's or above. Before we remove biosolids to holding tanks we remove extra supernate. We do have some digestion occur in our sludge holding tanks. After digestion in holding tanks we remove extra decant from tank to thicken biosolids. ☐ Check this box if you need additional space for your answer.	5. VECTOR ATTRACTION REDUCTION Which vector attraction reduction option(s) is met for the biosolids at your facility? Attach the appropriate certification statement(s) (Appendix C) and documentation to demonstrate compliance. ☐ Option 1 (minimum 38 percent reduction in volatile solids) ☐ Option 2 (Anaerobic process, with bench-scale demonstration) ☐ Option 3 (Aerobic process, with bench-scale demonstration) ☐ Option 4 (Specific oxygen uptake rate, aerobically digested) ☐ Option 5 (Aerobic process plus raised temperature) ☐ Option 6 (Raise pH to 12 and retain at 11.5) ☐ Option 7 (75% solids with no unstabilized solids) ☐ Option 8 (90% solids with unstabilized solids) ☒ Option 9 (Injection below land surface) ☐ Option 10 (Covering active sewage sludge unit daily)
2. PATHOGEN REDUCTION Which class of pathogen reduction does the biosolids meet at your facility? ☐ Class A - Continue with item 3. ☒ Class B - Continue with item 4.	6. Describe, on this form or another sheet of paper, any secondary modes used at your facility to meet vector attraction reduction requirements: Option 1 (Minimum 38 percent reduction in volatile solids) . ☐ Check this box if you need additional space for your answer.
3. CLASS A BIOSOLIDS If Class A, check the alternative used and attach the appropriate certification statement(s) (Appendix C) and documentation demonstrating compliance. ☐ Class A – Alternative 1 ☐ Class A – Alternative 2 ☐ Class A – Alternative 3 ☐ Class A – Alternative 4 ☐ Class A – Alternative 5 (indicate which PFRP) ☐ (a) composting ☐ (b) heat drying ☐ (c) heat treatment ☐ (d) thermophilic aerobic digestion ☐ (e) beta ray irradiation ☐ (f) gamma ray irradiation ☐ (g) pasteurization ☐ Class A – Alternative 6 (attach PFRP equivalent documentation)	7. RE-USE/DISPOSAL ALTERNATIVES Should the circumstances arise, describe alternative method(s) for the proper treatment, use or disposal of biosolids (include seeking authorization from the permitting authority). Haul to approved licensed landfill. If the need arose the city of cadillac would determine how to haul biosolids to a landfill. The cost of such hauling would determine treatment options such as belt filter press. etc. With the holding capacity currently the city would have ample time to dispose of biosolids safely. ☐ Check this box if you need additional space for your answer.
4. CLASS B BIOSOLIDS If Class B, check the alternative used. Attach the appropriate certification statement(s) (Appendix C) and documentation to demonstrate compliance. Note Class B Site Restrictions on Page 7. ☐ Class B –Alternative 1 ☐ Class B –Alternative 2 (indicate which PSRP) ☐ (a) aerobic digestion ☐ (b) air drying ☒ (c) anaerobic digestion ☐ (d) composting ☐ (e) lime stabilization (pH at 25° C or equivalent) ☐ Class B –Alternative 3 (attach PSRP equivalent documentation)	

RESIDUALS MANAGEMENT PROGRAM

SECTION III – LAND APPLICATION PLAN

The following Site Restrictions contained in R323.2414 (f)(I) to (viii) apply to all land-applied biosolids meeting the class B pathogen reduction requirements.

(f) All of the following provisions apply to site restrictions:

- (i) A land owner shall not harvest food crops that have harvested parts which touch the biosolids/soil mixture and which are totally above the land surface for 14 months after biosolids are applied.
- (ii) A land owner shall not harvest food crops that have harvested parts below the surface of the land for 20 months after biosolids are applied if the biosolids remain on the land surface for 4 months or longer before incorporation into the soil.
- (iii) A land owner shall not harvest food crops that have harvested parts below the surface of the land for 38 months after biosolids are applied if the biosolids remain on the land surface for less than 4 months before incorporation into the soil.
- (iv) A land owner shall not harvest food crops, feed crops, and fiber crops for 30 days after biosolids are applied.
- (v) A land owner shall not graze animals on the land for 30 days after biosolids are applied
- (vi) A land owner shall not harvest turf grown on land where biosolids are applied for 1 year after biosolids are applied if the harvested turf is placed on either land that has a high potential for public exposure or a lawn, unless otherwise specified by the permitting authority.
- (vii) A land owner shall restrict public access to land that has a high potential for public exposure for 1 year after biosolids are applied.
- (viii) A land owner shall restrict public access to land with a low potential for public exposure for 30 days after biosolids are applied.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****B. Procedures**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
1. ENVIRONMENTAL MANAGEMENT SYSTEM (EMS) Optional Would your facility possibly be interested in implementing a voluntary EMS and "Code of Good Practices" as administered by the National Biosolids Partnership? ☐ Yes -Information will be sent to you on implementing an EMS ☒ No	4. LAND APPLICATION METHOD Describe the methods utilized for land application. Sub Surface injection with an "A" Sludge Applicator. Liquid biosolids is injected beneath the surface with no significant amount present above ground. Application rates are set by equipment operators to meet amount of tons per acre as set by WasteWater supervisor. The rates shall be set to agromomic rates. ☐ Check this box if you need additional space for your answer.
2. TRANSPORTATION Describe the method of handling and transporting residuals from the treatment facility to the site of land application. We use a Quilified Sub Contractor to Haul Our Biosolids. Lowest Bid With Best References. Biosolids will be removed from the storage tanks with a portable hydrolic pump and pumped into 9000 gallon tanker trucks. All Tankers will be properly registered and pass annual federal inspections. When leaving facility trucks will use Plett road exit. Drivers will comply with all wieght restrictions that may be applicable on roads leading to application sites. Trucks will travel main roads if possible and secondary roads only when necessary. ☐ Check this box if you need additional space for your answer.	
3. SPILL PREVENTION / RESPONSE PLAN Describe your spill prevention measures and steps to be followed should a spill occur. 1. On site truck fill station used to fill all tankers. 2. Equipment operators starts filling tanker and when full shuts off pump and shuts valve on discharge pipes. Operator closes hacht on truck. 3. Safe driving rules are followed to the land application site. 4. At application site a proper staging area will be established for easy access to taker truck. A flat area to allow for diking should be chosen. 5. A double valve system with high pressure hose shall be used. both valves will be opened for filling applicator and closed when finished. Small Spills (less then 100 gallons) will be contained at fill site. sand will be applied and mixture will be hauled away. Large Spills Land application staff will respond immediately. Diking pumping and cleanup shall be used to control site. every effort will be made to impact as little of surronding area's as possible. ☐ Check this box if you need additional space for your answer.	5. APPLIER OVERSIGHT Describe the method of applier oversight. Include performance specifications and any training/certification requirements. See Appendix C for applier certification requirements. Get yearly reports with all certifications needed. Do site inspections during and after application of biosolids. The City of Cadillac will by its bidding of Biosolids get the most experienced and quilified contractor . This should minimize the potential for future problems. Prior to application the city will review all information on each proposed site to assure that the information on file is accurate and current. The city will review the most current laboratory analyses to assure compliance with all applicable rules and regulation. ☒ Check this box if you need additional space for your answer

RESIDUALS MANAGEMENT PROGRAM

SECTION III – LAND APPLICATION PLAN

B. Procedures (continued)

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WasteWater Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
6. REPORTING Describe your facility's reporting procedure (provide example forms, letters, etc. as an attachment). Specific reporting requirements are summarized in Appendix C. We send a report consisting of cover letter ,waste disposal reportss and lab results to district office. Send yearly biosolids to Michigan DEQ. City of Cadillac will make land packages that will contain owner permission to apply forms, site idenification forms ,soil analysis , an appropriate plat map , soil map of site, biosolids analyses, and notification letters. Waste disposal sheets will be completed when hauling is completed. Annual report is made THE City of Cadillac, as preparer of class B biosolids , will complete a certification statement for pathogen reduction and vector attraction. The applier will complete certifications for site management, site restrictions, and vector attraction reduction by injection. ☐ Check this box if you need additional space for your answer	8. COMPLAINT RESPONSE Describe the procedure for responding to citizen complaints. Include details of any biosolids educational programs. We try to get persons name and visit them and explain our process to them. We have started to test our biosolids and found tout that they meet Class "A" requirements. We try to educate and relieve the person that complains fears. We also give numorous tours during the year to educate the public. ☒ Check this box if you need additional space for your answer
7. RECORDKEEPING Describe your facility's recordkeeping procedures and retention schedules. Specific recordkeeping and certification requirements are summarized in Appendix C. We keep facilities report for a period of at least a minimum of five years. In most cases longer. The last five years are kept in supervisors office. The years before that are kept in plant archives. ☐ Check this box if you need additional space for your answer	9. SAMPLING PLAN a. Describe the proposed sample frequency and quality control methods utilized to assure representative samples are obtained. Minimum Sample Frequencies are provided on Page 10. We only need to sample once per year but we test before every hauling cycle. This means we test biosolids two times a year. We mix our biosolids for about a month before we haul.We have two ways of mixing. Usually we use air to mix by sinking aeration pipes to bottom off tank.If biosolids is to thick we will use hydraulic pump to mix.We take a sample from the tank over a 8 hour time to make composite. Currently we may need to sample quarterly. ☐ Check this box if you need additional space for your answer b. Describe the procedure for assuring that analytical results are below regulated limits and appropriate methods were used. Pollutant Limit Tables are provided on Page 11. We test our biosolids in house and we also send a split sample out to a commercial lab. We use our answers to confirm commercial labs results. We only use commercial lab results in our reports. The commercial lab sends confirmation of tests methods and QA/QC ☐ Check this box if you need additional space for your answer

SAMPLE FREQ

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****Minimum Sample Frequencies****R 323.2412 Frequency of monitoring.**

Table 7

Amount of Biosolids Produced (per 365-day period)		
English Dry Tons	Metric Dry Tons	Frequency
Greater than zero, but less than 320	Greater than zero, but less than 290	Once per year
Equal to or greater than 320, but less than 1,650	Equal to or greater than 290, but less than 1,500	Once per quarter (4 times per year)
Equal to or greater than 1,650, but less than 16,500	Equal to or greater than 1,500, but less than 15,000	Once per 60 days (6 times per year)
Equal to or greater than 16,500	Equal to or greater than 15,000	Once per month (12 times per year)

Note: Additional monitoring beyond that stated above may be required to obtain representative data of the material proposed to be land applied or to meet the following additional requirements of the Part 24 Rules. The following words in bold below are not part of the Part 24 Rules.

(4) If biosolids are accumulated before removal, the monitoring frequency, at a minimum, is that specified in this rule. If monitoring of biosolids or a derivative indicates a pollutant concentration in excess of that provided in table 3 of R 323.2409(5)(c) (**see page 11 of this document**), then the monitoring frequency shall be increased to not less than twice that provided for in table 7 (**sample frequency table above**) until pollutant concentrations are at or below the concentrations provided in table 3 of R 323.2409(5)(c). In the case of biosolids accumulating for periods of more than 1 year, biosolids must be monitored at the frequency determined in table 7 only in the year the biosolids are used. For a generator who removes biosolids monthly or more frequently, monitoring is required at least once per month each month that the biosolids are removed, unless more frequent monitoring is required in subrule(1) of this rule.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****R 323.2409 APPLICATION OF LIMITATIONS**

(a) TABLE 1 -- Ceiling Pollutant Concentrations

Pollutant	Ceiling Concentration (milligrams per kilogram) ¹ (dry weight basis)
Arsenic	75
Cadmium	85
Copper	4300
Lead	840
Mercury	57
Molybdenum	75
Nickel	420
Selenium	100
Zinc	7500

(b) TABLE 2 -- Cumulative Pollutant Loading Rates

Pollutant	Cumulative Pollutant Loading Rate	
	kilograms per hectare	pounds per acre
Arsenic	41	37
Cadmium	39	35
Copper	1500	1335
Lead	300	267
Mercury	17	15
Nickel	420	374
Selenium	100	89
Zinc	2800	2492

(c) TABLE 3 -- Pollutant Concentrations

Pollutant	Concentration (milligrams per kilogram) (on a dry weight basis)
Arsenic	41
Cadmium	39
Copper	1500
Lead	300
Mercury	17
Nickel	420
Selenium	100
Zinc	2800

(d) TABLE 4 -- Annual Pollutant Loading Rates

Pollutant	Annual Pollutant Loading Rate ¹	
	kilograms per hectare	pounds per acre
Arsenic	2.0	1.8
Cadmium	1.9	1.7
Copper	75	67
Lead	15	13
Mercury	0.85	0.76
Nickel	21	19
Selenium	5.0	4.5
Zinc	140	125

¹ per 365-day period.

[illegible]

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****C. Site Information (continued)****PLEASE TYPE OR PRINT**

FACILITY NAME Cadillac WasteWater Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
4. NEW SITES a. Indicate the geographical area covered by your Land Application Plan. Wexford and Missaukee counties ☐ Check this box if you need additional space for your answer b. Describe your site selection criteria. Note the Management Practices summary on Page 14 and Class B site restrictions - Page 7. Applier/Preparer certification requirements are summarized in Appendix C. We try to pick land closest to wastewater plant that is out in the farming areas around Cadillac. We try to find land that meet all of the site requirements so that there can be nodoubt that it is a compliant site. we look for site size, ease of access, slope of site, soil conditions and farmers crop rotation. ☐ Check this box if you need additional space for your answer	7. AGRONOMIC RATE a. Will biosolids be applied at the agronomic rate for the crop to be grown? See Appendix E for agronomic rate calculations. ☒ Yes - Continue with Item 8. ☐ No - Biosolids must be applied at the agronomic rate unless done in accordance with a DEQ approved Site Reclamation Plan. *NOTE: Be advised that application rates may also need to be further restricted for operational reasons or based on loading rates for biosolids exceeding Table 3 limits or to avoid exceeding the 300 lb/ac P limit in site soils. Refer to SWQD guidance for calculations of agronomic rates based on Phosphorus.
5. LAND OWNER AGREEMENTS Do you have a signed landowner agreement for each existing site? See Appendix D for Landowner Agreement considerations. ☒ Yes - Continue with Item 6. ☐ No - Rule 323.2413(2)(b) and (c) requires written consent to apply biosolids and a written agreement not to apply biosolids from other sources or septage to a listed land application site.	8. SITE NOTIFICATION Specific notification requirements and a sample public notification letter are provided in Appendix F. A DEQ site identification form is provided on page 15. Optional: Describe your facility's site notification procedure, including informal DEQ notification prior to seasonal or wintertime application. Provide any other sample forms and letters. See samples ☐ Check this box if you need additional space for your answer
6. OUT OF STATE SITES Will the proposed land application activity occur on lands outside of the State of Michigan? ☐ Yes (describe procedure for permitting authority notification). ☒ No, Continue with Item 7. ☐ Check this box if you need additional space for your answer	

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****MANAGEMENT PRACTICES****Slope Restrictions R 323.2410(4)**

A person shall not apply bulk biosolids on lands having a slope of more than 6% for surface application or more than 12% for subsurface injected biosolids, unless the person uses the bulk biosolids in accordance with a DEQ-approved site management plan.

Isolation Distance Requirements R 323.2410(11)

TABLE 6

ISOLATION DISTANCE REQUIREMENTS		
Isolation from existing:	Distance (feet)	
	Injection or Surface application with incorporation*	Surface application without incorporation
Municipal well (type I or type IIA)**	2000	2000
Non-community public water supply (type IIB or type III)	800	800
Domestic well	100	150
Homes	100	150
Commercial Buildings	100	150
Surface waters***	50	150

* Incorporation must be within 48 hours, unless a shorter time period is specified in these rules.

** As defined and specified in Act No. 399 of the Public Acts of 1976, as amended, being §325.1001 et seq. of the Michigan Compiled Laws and known as the safe drinking water act. As specified in Act No. 399 of Public Acts of 1976, as amended, the term includes water supplies such as schools, restaurants, industries, campgrounds, parks, and hotels.

*** Surface waters do not include grassed drainage ways or drainage ways that are tilled and planted.

Refer to Rule R 323.2410 for the entire management practices section of Part 24 Rules.

DEQ SITE IDENTIFICATION FORM

Complete the following information and provide the attachments listed below for each new site or existing site that has not had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e. new land owner).

FACILITY NAME Cadillac WasteWater Treatment Plant		NPDES or COC PERMIT NUMBER MI0020257	
---	--	--	--

SITE IDENTIFICATION FORM

A. Site ID

B. County/Township:

C. State Planar Coordinates:

D. Lat/Long (nearest second):

E. Site Ownership

F. Site Characteristics

Site ID #		Street or Route #		
County			Township	
1/4	1/4	Section	Town	Range
Latitude		Longitude		Method Used (i.e. GPS, topo)

Name of Owner		Phone	
Street or P.O. Box			
City or Town		State	Zip
Is there a written agreement from the owner to land apply? ☐ Yes ☐ No, written agreement is required		Previous landowners name Property purchase date	

Name of Lessee (if different)		Phone	
Street or P.O. Box			
City or Town		State	Zip

Site Type ☐ Agricultural ☐ * Reclamation Site ☐ Forest ☐ ** Lawn/Home Garden *Separate approval is required **Requires EQ Biosolids		Soils Information Date of last soils analysis _____ *Phosphorous _____ units _____ Potassium _____ units _____ pH _____ Soil type(s) _____ *300 lb/ac Bray P1 maximum									
Expected crops/vegetation to be grown <table style="width: 100%;"> <tr> <th style="width: 40%;">Crop/Vegetation</th> <th style="width: 60%;">Nitrogen Requirement</th> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> </table>		Crop/Vegetation	Nitrogen Requirement	_____	_____ lbs. N/acre	_____	_____ lbs. N/acre	_____	_____ lbs. N/acre	Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☐ *Higher (Requires an approved site reclamation plan)	
Crop/Vegetation	Nitrogen Requirement										
_____	_____ lbs. N/acre										
_____	_____ lbs. N/acre										
_____	_____ lbs. N/acre										
Public Exposure Is this land with a high potential for public exposure? (See Definitions section on Page 19) ☒ Yes ☐ No		If agricultural, is the site tiled? ☐ Yes ☐ No ☐ NA									

On an attached sheet provide a copy of the following:

- A Farm Services Agency map (formerly ASCS) of the proposed application site with property boundaries, surface waters, and discharge location of field tiles when present.
- A copy of the soil test analysis that is no older than 2 years old at the time of land application.
- A copy of the plat map (or other map showing property dimensions) with the proposed land application site highlighted.
- A soil survey map with the land application site identified.

DEQ SITE IDENTIFICATION FORM

Complete the following information and provide the attachments listed below for each new site or existing site that has not had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e. new land owner).

Land Application Site List

Provide the following information for every new or existing site that you intend to use in the next five years (biosolids permit cycle). Each listed site should either have had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998, or the completed forms should be included with this document. To add new sites to the Land Application Site List during the biosolids permit cycle, modify your RMP by submitting a completed DEQ Site Identification Form with attachments, a copy of your public notification letter and wait the required ten-day notification period. Use additional sheets as necessary.

PLEASE TYPE OR PRINT

Facility Name Cadillac WasteWater Treatment Plant				NPDES or COC PERMIT NUMBER MI0020257		
Site Identification Number	Latitude	Longitude	Acres	Owners Last Name	New Site? (check if yes)	CPLR Site? (check if yes)
IV01	N44 14.567'	W85 17.589'	33	VANDERVLUGHT	☐	☐
IV02	N44 14.429'	W85 17.414'	18	VANDERVLUGHT	☐	☐
IV03	N44 14.308	W85 17.589	37	VANDERVLUGHT	☐	☐
IV04	N44 14.00	W85 16.900	16	VANDERVLUGHT	☐	☐
IV05	N44 13.940	W85 16.900	16	VANDERVLUGHT	☐	☐
IV06	N44 13.877	W85 16.900	16	VANDERVLUGHT	☐	☐
IV07	N44 13.811	W85 16.900	14	VANDERVLUGHT	☐	☐
IV08	N44 13.735	W85 16.900	15	VANDERVLUGHT	☐	☐
IV09	N44 14.030	W85 16.550	8	VANDERVLUGHT	☐	☐
IV10	N44 13.940	W85 16.471	7	VANDERVLUGHT	☐	☐
IV11	N44 14.429	W85 17.360	12	VANDERVLUGHT	☐	☐
IV12	N44 14.429	W85 17.300	6	VANDERVLUGHT	☐	☐
IV13	N44 14.429	W85 14.240	6	VANDERVLUGHT	☐	☐
IV14	N44 14.429	W85 17.625	8	VANDERVLUGHT	☐	☐
IV15	N44 14.279	W85 17.035	6	VANDERVLUGHT	☐	☐
KL01	N44 14.279	W85 17.035	46	LIEDEKE	☐	☐
CB01	N44 14.279	W85 15.649	25	BALDWIN	☐	☐
CB02	N44 15.696	W85 15.579	23	BALDWIN	☐	☐
					☐	☐
					☐	☐
					☐	☐

E. Biosolids Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Wastewater Treatment Plant	NPDES PERMIT NUMBER MI0020257
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6. Land Application Site List

Provide the following information for every new or existing site that may be used in the next five years (biosolids permit cycle). Each listed site should either have had a Site Identification Form with attachments submitted to the MDEQ since January 1, 1998, or that information should be included with this form. Additional sites may be added to the Land Application Site List during the biosolids permit cycle by submitting a completed Site Identification Form with the appropriate attachments and waiting the required ten day notification period. Use additional pages as necessary.

Site Identification Number	Latitude (dd,mm,ss)	Longitude (dd,mm,ss)	Acres	Owners Last Name	New Site? (check if yes)	CPLR* Site? (check if yes)
JE 001	N44 17.241	W085 13.274	40	EUBANK	☒	☐
WE 001	N44 17.340	W085 13.173	40	EUBANK	☒	☐
LH 001	N44 16.911	W085 14.941	34	HARRINGTON	☒	☐
LH 002	N44 16.930	W085 14.805	18	HARRINGTON	☒	☐
LH 003	N44 16.911	W085 13.607	16	HARRINGTON	☒	☐
LH 004	N44 17.061	W085 13.591	6.5	HARRINGTON	☒	☐
PM 001	N44 11.208	W085 20.372	7	MILLER	☒	☐
PM 002	N44 11.294	W085 20.372	7	MILLER	☒	☐
PM 003	N44 11.350	W085 20.372	10	MILLER	☒	☐
PM 004	N44 11.376	W085 20.372	10	MILLER	☒	☐
PM 005	N44 11.415	W085 20.372	10	MILLER	☒	☐
PM 006	N44 11.380	W085 20.802	25	MILLER	☒	☐
					☐	☐
					☐	☐
					☐	☐
					☐	☐
					☐	☐
					☐	☐
					☐	☐
					☐	☐
					☐	☐

* see definitions

RESIDUALS MANAGEMENT PROGRAM

SECTION IV – SIGNATURE PAGE

PLEASE TYPE OR PRINT

Facility Name

Cadillac Wastewater Treatment Plant

NPDES or COC PERMIT NUMBER

MI0020257

1. CERTIFICATION

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with the system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name and official title

Jeffrey M. Dietlin
Water Resources Supervisor

Signature



Telephone number

231-775-2841

Date signed

04/15/2002

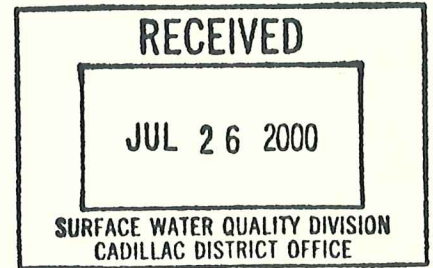
Upon request from the State, you may be required to submit additional information necessary to assess biosolids use or disposal practices at your facility or to identify appropriate permitting requirements.

SEND COMPLETED FORMS TO THE APPROPRIATE DISTRICT OFFICE LISTED IN APPENDIX A.



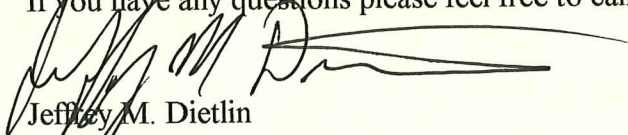
July 26 , 2000

Mr. Mike Person
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac , Michigan 49601



Enclosed is the RMP report for biosolids that was due on July 7 , 2000. I hope everything is in order. If you need any additional information please contact me.

If you have any questions please feel free to call me.


Jeffrey M. Dietlin
Water Resources Supervisor
Phone # : (231) 775 - 2841



State of Michigan Biosolids Land Application Program

Facility Name
City of Cadillac Waste Water Treatment Plant

Residuals Management Program Development Document

General Instructions	Page	i
Section I - General Information	Pages	1 - 2
Section II - Biosolids Characterization	Pages	3 - 4
Section III - Land Application Plan	Pages	5 - 16
Signature Page	Page	17
Definitions	Pages	18 - 20
Additional Information	Page	21
Appendix	Pages	22 - 41



**SURFACE WATER QUALITY DIVISION
MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY**

RMP
EQ
Class A

PURPOSE

The Michigan Biosolids Program establishes standards for the land application and beneficial recycling of biosolids in the state. Any treatment works treating domestic sewage proposing to land apply biosolids in the state shall prepare the enclosed Residuals Management Program (RMP) Guidance Document to obtain authorization. Authorization is required under the auspices of either an individual NPDES Permit or General Permit "Certificate of Coverage" (COC), dependent upon the type of permit held by the generating facility proposing to land apply.

For assistance on completing any section in this document you may contact the appropriate DEQ Surface Water Quality Division (SWQD) Office listed in Appendix A.

Industrial or commercial facilities with process wastewater entering the waste stream do not qualify for issuance of a biosolids land application permit. Contact the DEQ Waste Management Division in the appropriate district office for information on handling commercial or industrial wastes.

AUTHORITY

The Part 24 Rules, Land Application of Biosolids, of Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended, (NREPA) provide authority to issue Biosolids Land Application Permits. Completion and subsequent approval of this document meets the RMP requirements contained in Rule 2403, Land Application Permit of the Part 24 Rules. Failure to prepare and submit an RMP prior to land applying biosolids is a violation of Part 24 Rules and may result in penalties for such violation.

Rule 2403.

- (1) A generator or distributor shall have a valid permit before commencing any biosolids land application or distribution activity in the state of Michigan.
- (2) A generator or distributor shall submit a permit application on a form provided by the DEQ not less than 180 days before expiration of an existing permit, commencement of biosolids land application, or distribution of a biosolids derivative, except at retail.
- (3) All of the following provisions apply to a residuals management program:
 - (a) A generator shall submit a residuals management program for approval by the DEQ as required by its permit.
 - (b) A residuals management program submitted to the DEQ shall include all of the following information:
 - (i) Size and type of generating facility.
 - (ii) One year of records representing the volume and concentrations of pollutants in the biosolids.
 - (iii) Treatment process origin, for example, primary or secondary treatment and the volume of biosolids generated from each process.
 - (iv) A description of the treatment processes.
 - (v) Storage volume.
 - (vi) Transportation methods and spill prevention plan.
 - (vii) Land application method.
 - (viii) Land application site list.
 - (ix) Land application plan.
 - (x) Pathogen reduction method.
 - (xi) Vector attraction reduction method.
 - (xii) Monitoring program.
 - (c) Upon approval by the DEQ, the generating facility shall implement the approved residuals management program.
 - (d) A generating facility may modify the approved residuals management program by submitting a proposed modification to the DEQ for approval. The modification shall become effective upon approval by the DEQ.
- (4) A person shall land apply biosolids or prepare biosolids for land application in accordance with the requirements established in these rules.

The Michigan Department of Environmental Quality (DEQ) will not discriminate against any individual or group on the basis of race, sex, religion, age, national origin, color, marital status, disability, or political beliefs. Questions or concerns should be directed to the Office of Personnel Services, PO Box 30473, Lansing MI 48909

RESIDUALS MANAGEMENT PROGRAM**SECTION I – GENERAL INFORMATION**

PLEASE TYPE OR PRINT

1. NPDES or COC NUMBER MI0020256			4. FACILITY MAILING ADDRESS Street Address or P.O. Box (or check box to use address corresponding to item number ☒ 2 ☐ 3)																				
2. GENERATOR NAME AND MAILING ADDRESS Name <u>City of Cadillac WasteWater Treatment Plant</u> Superintendent, CEO, Principal Executive Officer Name <u>Jeffrey M. Dietlin</u> Street Address or P.O. Box <u>200 North Lake Street</u>			Additional Street Address or P.O. Box Information																				
City or Village <u>Cadillac</u>	State <u>MI</u>	ZIP Code <u>49601</u>	City or Village State ZIP Code																				
Telephone (include area code) <u>231-775-2841</u>			5. BIOSOLIDS CONTACT PERSON Name (if different than Item 2) Title																				
3. FACILITY NAME AND LOCATION (if different) Name <u>Cadillac Waste Water Treatment Plant</u> Street Address <u>1121 Plett Road</u> Additional Street Address			ADDRESS Street Address or P.O. Box (or check box to use address corresponding to item number ☒ 2 ☐ 3 ☐ 4)																				
City or Village <u>Cadillac</u>	State <u>MI</u>	ZIP Code <u>49601</u>	City or Village State ZIP Code																				
Latitude (nearest 15 seconds) Longitude			Telephone (include area code)																				
Telephone (include area code) <u>231-775-2841</u>			Fax Number (with area code)																				
Fax Number (include area code) <u>231-775-2876</u>			6. Type of Treatment Facility Publicly owned treatment works ☒ Privately owned treatment works ☐ Federally owned treatment works ☐ Blending or treatment operation ☐ State owned treatment works ☐ Tribal owned treatment works ☐ Other ☐ explain:																				
Email Address (if any) <u>wwtp@michweb.net</u>																							
7. LAND APPLICATION REPORT FORMS Check the box that corresponds to the address (above) to which forms should be mailed. ☒ 2 – Applicant Name & Mailing Address ☐ 3 - Facility Name & Location ☐ 4 - Facility Mailing Address ☐ 5 - Contact Person SEND FORMS TO THE ATTENTION OF: <u>Jeffrey M. Dietlin</u>																							
8. Provide the names, addresses and telephone numbers of the Land Application Contractors used by the facility, if applicable. <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 30%;">Name of Contractor:</td> <td><u>Synagro Midwest</u></td> </tr> <tr> <td>Address:</td> <td><u>323 Martindale</u></td> </tr> <tr> <td>City, State, ZipCode:</td> <td><u>Sparta , MI 49345</u></td> </tr> <tr> <td>Telephone:</td> <td><u>616-887-1144</u></td> </tr> <tr><td colspan="2"> </td></tr> <tr> <td>Name of Contractor:</td> <td></td> </tr> <tr> <td>Address:</td> <td></td> </tr> <tr> <td>City, State, ZipCode:</td> <td></td> </tr> <tr> <td>Telephone:</td> <td></td> </tr> </table>						Name of Contractor:	<u>Synagro Midwest</u>	Address:	<u>323 Martindale</u>	City, State, ZipCode:	<u>Sparta , MI 49345</u>	Telephone:	<u>616-887-1144</u>			Name of Contractor:		Address:		City, State, ZipCode:		Telephone:	
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Name of Contractor:																							
Address:																							
City, State, ZipCode:																							
Telephone:																							

RESIDUALS MANAGEMENT PROGRAM**SECTION I – GENERAL INFORMATION**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
9. RESIDUALS HANDLING Provide total English dry tons per 365-day period of biosolids handled under the following practices: Amount generated at the facility: <u>250</u> Amount received from off site: <u>0</u> Amount treated on site (including blending): <u>250</u> Amount sold or given away in a bag or other container for application to the land: <u>0</u> Amount of bulk biosolids shipped off site for treatment or for sale/give-away in a bag or other container for application to the land: <u>0</u> Amount applied to land in bulk form: <u>250</u> Amount fired in biosolids incinerator: <u>0</u> Amount sent to municipal solid waste landfill: <u>0</u> Amount used or disposed by another practice: <u>0</u> Describe: _____ No Answer 13, 14, 15	10. RESIDUALS STORAGE Enter the volume of residual storage capacity at this facility. 1.6 ☒ million gallons or ☐ cubic feet 11. WASTEWATER FLOW RATE Provide design and actual flow rates in million gallons per day (MGD). Design flow <u>3.2</u> MGD Average Actual Flow <u>2.3</u> MGD 12. SERVICE AREA Enter the population serviced or NA if not applicable. <u>11500</u> 13. INDUSTRIAL PRETREATMENT PROGRAM Does the generating facility have, or are they required to have, an Industrial Pretreatment Program? (See Appendix B) Yes – On attached sheet provide list of Significant Industrial Users and list of pollutants monitored. No – Continue with item 14. 14. SEPTAGE Does your facility accept septage? Yes – Estimate amount received monthly _____ gal. No – Continue with item 15. 15. BLENDING AND TREATMENT FACILITIES Are (or will) biosolids from another facility (not including septage) be sent to your facility for treatment (including blending) or placement in a bag or other container for sale or give-away? Yes – Contact the Pretreatment and Biosolids Unit, Lansing No – Continue with item 16.
16. SOLIDS FLOW DIAGRAM, NARRATIVE DESCRIPTION, and PRODUCTION DATA Provide a flow diagram (using 8 1/2" x 11" paper if possible) showing the solids handling process through the facility. Identify all treatment units and sampling points. Please attach this diagram. Include a narrative description of all solids handling through the facility and provide solids production data of solids unit processes. Provide detailed information on residuals stabilization process including all solids handling unit processes (i.e. collection, dewatering, digestion, chemical feeds, storage). Include a narrative that briefly describes the history of the residuals handling practices at the treatment facility and any future plans for upgrade. OPTIONAL- To assist in calculating production data for your facility, Excel spreadsheets of various processes have been developed by the DEQ Operator Training Unit and are available from the appropriate DEQ SWQD Office listed in Appendix A or via the biosolids homepage at http://www.deq.state.mi.us/swq/ . If you are an industrial or commercial facility treating domestic sewage sludge only, the line diagram shall include all operations contributing wastewater including process and production areas, sanitary flows, cooling water and storm water runoff. Include a narrative, which provides a brief description of the manufacturing processes.	

RESIDUALS MANAGEMENT PROGRAM**SECTION II – BIOSOLIDS CHARACTERIZATION**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant				NPDES or COC PERMIT NUMBER MI10020257			
1. BIOSOLIDS CHARACTERISTICS Report one year biosolids monitoring data, and in no case less than three sampling events for the following Part 24 rules required parameters. Provide the actual analytical data sheets as an attachment. All sampling shall be representative of the biosolids being proposed to be applied to the land and in accordance with the minimum sample frequencies provided on the Page 10. Analytical methods shall be in accordance with R 323.2406 (2) Methods for Biosolids. Data that is available for parameters not specifically listed on this page shall be provided in the Pollutants of Concern table on page 4, Item 3 of this section.							
Parameter	Average Monthly Concentration	Maximum Monthly Concentration	Units	Number of Analyses	Method Detection Limit	Test Method	Sample Type
Inorganics							
Total Solids	4.97	7.8	%	3		SM2540B	☐ Grab ☒ Composite
Total Arsenic	20.4	25.2	Mg/kg	3		EPA 6020	☐ Grab ☒ Composite
Total Cadmium	8.18	9.7	Mg/kg	3		EPA 6020	☐ Grab ☒ Composite
Total Copper	900	981	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Lead	135.3	142	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Mercury	1	1.4	Mg/kg	3		EPA7471A	☐ Grab ☒ Composite
Total Molybdenum	60.67	75	Mg/kg	3		EPA6010B	☐ Grab ☒ Composite
Total Nickel	48.3	57	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Total Selenium	2.77	6.5	Mg/kg	3		EPA 7740	☐ Grab ☒ Composite
Total Zinc	1933.33	2090	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite
Nutrients							
Total Kjeldahl Nitrogen	5.27	5.5	%	3		EPA 351.2	☐ Grab ☒ Composite
Ammonium Nitrogen	2.6	2.6	%	3		EPA 350.1	☐ Grab ☒ Composite
Total Phosphorus	4.83	6.3	%	3		SM4500-PE	☐ Grab ☒ Composite
Total Potassium	2233.33	2470	Mg/kg	3		EPA 6010B	☐ Grab ☒ Composite

As Higher than expected
 - Heavy @ Creek
 Zone Higher than normal

RESIDUALS MANAGEMENT PROGRAM**SECTION II – BIOSOLIDS CHARACTERIZATION**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
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2. Pollutants of Concern

Do you believe or suspect that pollutants (other than parameters listed on the previous page) could be present in the biosolids to be land applied at concentrations that could possibly make them unsuitable for land application?

☐ Yes - Describe the circumstances below and provide representative analytical data for those pollutants of concern in Item 3.

☒ No

3. Report any biosolids monitoring data from the last permit cycle for parameters not specifically listed on the previous page. Include the actual analytical data sheets as an attachment. Upon submittal review, additional monitoring may be required if SWQD has reason(s) to suspect that the information provided (or not provided) does not adequately characterize the residuals proposed to be land applied. For assistance on completing this section or determining the necessity for completing this section, you may contact the appropriate DEQ SWQD office (see Appendix A).

Parameter	Average Concentration	Maximum Concentration	Units	Number of Analyses	Method Detection Limit	Test Method	Sample Type
ALUMINUM	7116.7	7690	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
CALCIUM	32633	35100	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
CHROMIUM	125	143	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
MAGNESIUM	3978	4170	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
SILVER	63.3	80.9	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
SODIUM	3937	4150	mg/kg	3		EPA6010B	☐ Grab ☒ Composite
CHLORIDE	4377	4480	mg/kg	3		EPA300A	☐ Grab ☒ Composite
SULFATE	675	1310	mg/kg	3		EPA300A	☐ Grab ☒ Composite
BIS (2-ETHYLHEXYL) PHTHALATE	62000	64000	ug/kg	2		EPA8270	☐ Grab ☒ Composite
METHYLENE CHLORIDE	1400	2800	ug/kg	2		EPA8270	☐ Grab ☒ Composite
PRIORITY POLLUTANTS	BELOW DET.	BELOW DET.	ug/kg	2		EPA8270	☐ Grab ☒ Composite
							☐ Grab ☒ Composite
							☐ Grab ☒ Composite
							☐ Grab ☒ Composite

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****A. Biosolids Treatment****PLEASE TYPE OR PRINT**

FACILITY NAME Cadillac Was	NPDES or COC PERMIT NUMBER MI0020257
1. PATHOGEN / VECTOR ATTRACTION REDUCTION Describe, on this form or another sheet of paper, any treatment processes used at your facility to reduce pathogen and vector attraction properties in biosolids: Anaerobic and With sludge holding tanks ☐ Check this box if you need additional space for your answer.	5. VECTOR ATTRACTION REDUCTION Which vector attraction reduction option(s) is met for the biosolids at your facility? Attach the appropriate certification statement(s) (Appendix C) and documentation to demonstrate compliance. ☐ Option 1 (minimum 38 percent reduction in volatile solids) ☐ Option 2 (Anaerobic process, with bench-scale demonstration) ☐ Option 3 (Aerobic process, with bench-scale demonstration) ☐ Option 4 (Specific oxygen uptake rate, aerobically digested) ☐ Option 5 (Aerobic process plus raised temperature) ☐ Option 6 (Raise pH to 12 and retain at 11.5) ☐ Option 7 (75% solids with no unstabilized solids) ☐ Option 8 (90% solids with unstabilized solids) ☒ Option 9 (Injection below land surface) ☐ Option 10 (Covering active sewage sludge unit daily)
2. PATHOGEN REDUCTION Which class of pathogen reduction does the biosolids meet at your facility? ☐ Class A - Continue with item 3. ☒ Class B - Continue with item 4.	6. Describe, on this form or another sheet of paper, any secondary modes used at your facility to meet vector attraction reduction requirements: OPTION 1 (MINIMUM 38 PERCENT REDUCTION IN VOLATILE SOLIDS) ☐ Check this box if you need additional space for your answer.
3. CLASS A BIOSOLIDS If Class A, check the alternative used and attach the appropriate certification statement(s) (Appendix C) and documentation demonstrating compliance. ☐ Class A – Alternative 1 ☐ Class A – Alternative 2 ☐ Class A – Alternative 3 ☐ Class A – Alternative 4 ☐ Class A – Alternative 5 (indicate which PFRP) ☒ (a) composting ☒ (b) heat drying ☒ (c) heat treatment ☒ (d) thermophilic aerobic digestion ☒ (e) beta ray irradiation ☒ (f) gamma ray irradiation ☒ (g) pasteurization ☐ Class A – Alternative 6 (attach PFRP equivalent documentation)	SEE Perm 120A RE 5 QUESTION 3? 1 PSRP WHY COMPOSTING AEROB?
4. CLASS B BIOSOLIDS If Class B, check the alternative used. Attach the appropriate certification statement(s) (Appendix C) and documentation to demonstrate compliance. Note Class B Site Restrictions on Page 7. ☐ Class B –Alternative 1 ☐ Class B –Alternative 2 (indicate which PSRP) ☒ (a) aerobic digestion ☐ (b) air drying ☐ (c) anaerobic digestion ☒ (d) composting ☐ (e) lime stabilization (pH at 25° C or equivalent)	7. RE-USE/DISPOSAL ALTERNATIVES Should the circumstances arise, describe alternative method(s) for the proper treatment, use or disposal of biosolids (include seeking authorization from the permitting authority). Haul to Approved Liscensed Landfill. ☐ Check this box if you need additional space for your answer.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****B. Procedures**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
1. ENVIRONMENTAL MANAGEMENT SYSTEM (EMS) Optional Would your facility possibly be interested in implementing a voluntary EMS and "Code of Good Practices" as administered by the National Biosolids Partnership? ☐ Yes -Information will be sent to you on implementing an EMS ☒ No	4. LAND APPLICATION METHOD Describe the methods utilized for land application. Sub Surface injection with an "A" Sludge Applicator ☐ Check this box if you need additional space for your answer.
2. TRANSPORTATION Describe the method of handling and transporting residuals from the treatment facility to the site of land application. We Use a Quilified Sub Contractor to Haul Our Sludge. Lowest Bid with Best References. ☐ Check this box if you need additional space for your answer.	5. APPLIER OVERSIGHT Describe the method of applier oversight. Include performance specifications and any training/certification requirements. See Appendix C for applier certification requirements. Get yearly report with all certificatioons needed. Do site inspections during and after application of biosolids. ☐ Check this box if you need additional space for your answer
3. SPILL PREVENTION / RESPONSE PLAN Describe your spill prevention measures and steps to be followed should a spill occur. Contain and Cleanup spills. We Will use appropriate equipment and manpower to clean up spills fast and efficiently. ☐ Check this box if you need additional space for your answer.	

Check PG 3 Permit
3 SAMPLES

See Contingency Plan
Permit PG 5 3 SAMPLES

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****B. Procedures (continued)**

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
6. REPORTING Describe your facility's reporting procedure (provide example forms, letters, etc. as an attachment). Specific reporting requirements are summarized in Appendix C. We send a report consisting of coverletterwaste disposal sheets and lab results to district office. Send yearly biosolids to EPA region 5. ☐ Check this box if you need additional space for your answer WASTE DISP SHEET NOT NECESSARY YOUNG REPORT TO DEC	8. COMPLAINT RESPONSE Describe the procedure for responding to citizen complaints. Include details of any biosolids educational programs. We try to get persons name and visit them and explain our process to them. We resrntly have tested our biosolids and found out that they met Class "A" requirements. We try to educate and relieve the persons tension on this subject. ☐ Check this box if you need additional space for your answer
7. RECORDKEEPING Describe your facility's recordkeeping procedures and retention schedules. Specific recordkeeping and certification requirements are summarized in Appendix C. We keep facilities report for a period of at least a minimum of five years. In most cases longer. The last five years are kept in supervisors office . The year before that are kept in our archives. ☐ Check this box if you need additional space for your answer CPLRS KORT UNDEF.	9. SAMPLING PLAN a. Describe the proposed sample frequency and quality control methods utilized to assure representative samples are obtained. Minimum Sample Frequencies are provided on Page 10. We only need to sample once per year but we test before every hauling cycle. This usually means we test oooooour biosolids 2 times a year. We mix our biosolids about a month before we haul and send out sample so that we have results before we haul. ☐ Check this box if you need additional space for your answer b. Describe the procedure for assuring that analytical results are below regulated limits and appropriate methods were used. Pollutant Limit Tables are provided on Page 11. We test our biosolids in house and we also send a split sample out to a commercial lab. We use our answers to confirm commercial labs results. We only use commercial labs test results in our repors. The commercial lab sends conformation of tests methods and QA/QC. ☐ Check this box if you need additional space for your answer WHORE ARE SAMPLES COLLECTED, HOW LONG MIXED BEFORE COLLECT HOW MANY SAMPLES TO COMPOSITE

[illegible]

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****C. Site Information (continued)****PLEASE TYPE OR PRINT**

FACILITY NAME Cadillac Waste Water treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
4. NEW SITES a. Indicate the geographical area covered by your Land Application Plan. Wexford and Missaukee counties ☐ Check this box if you need additional space for your answer b. Describe your site selection criteria. Note the Management Practices summary on Page 14 and Class B site restrictions - Page 7. Applier/Preparer certification requirements are summarized in Appendix C. We try to pick land closest to wastewater plant that is out in the farming areas around Cadillac. Try to findf land that mets all of the site requirement so well that there can be no doubt thast it is a compliant site. ☐ Check this box if you need additional space for your answer	7. AGRONOMIC RATE a. Will biosolids be applied at the agronomic rate for the crop to be grown? See Appendix E for agronomic rate calculations. ☒ Yes - Continue with Item 8. ☐ No - Biosolids must be applied at the agronomic rate unless done in accordance with a DEQ approved Site Reclamation Plan. *NOTE: Be advised that application rates may also need to be further restricted for operational reasons or based on loading rates for biosolids exceeding Table 3 limits or to avoid exceeding the 300 lb/ac P limit in site soils. Refer to SWQD guidance for calculations of agronomic rates based on Phosphorus.
5. LAND OWNER AGREEMENTS Do you have a signed landowner agreement for each existing site? See Appendix D for Landowner Agreement considerations. ☒ Yes - Continue with Item 6. ☐ No - Rule 323.2413(2)(b) and (c) requires written consent to apply biosolids and a written agreement not to apply biosolids from other sources or septage to a listed land application site.	8. SITE NOTIFICATION Specific notification requirements and a sample public notification letter are provided in Appendix F. A DEQ site identification form is provided on page 15. Optional: Describe your facility's site notification procedure, including informal DEQ notification prior to seasonal or wintertime application. Provide any other sample forms and letters. ☐ Check this box if you need additional space for your answer CHECK FOR VERBAGE
6. OUT OF STATE SITES Will the proposed land application activity occur on lands outside of the State of Michigan? ☐ Yes (describe procedure for permitting authority notification). ☒ No, Continue with Item 7. ☐ Check this box if you need additional space for your answer	

Michigan Department of Environmental Quality- Surface Water Quality Division
DEQ SITE IDENTIFICATION FORM

Complete the following information and provide the attachments listed below for each new site or existing site that has not had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e. new land owner).

FACILITY NAME Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
--	--

SITE IDENTIFICATION FORM

A. Site ID

Site ID #	Street or Route #			
County			Township	
1/4	1/4	Section	Town	Range
Latitude		Longitude		Method Used (i.e. GPS, topo)

B. County/Township:

C. State Planar Coordinates:

D. Lat/Long (nearest second):

E. Site Ownership

Name of Owner		Phone	
Street or P.O. Box			
City or Town		State	Zip
Is there a written agreement from the owner to land apply? ☐ Yes ☐ No, written agreement is required		Previous landowners name	
		Property purchase date	

Name of Lessee (if different)		Phone	
Street or P.O. Box			
City or Town		State	Zip

F. Site Characteristics

Site Type ☐ Agricultural ☐ *Reclamation Site ☐ Forest ☐ **Lawn/Home Garden *Separate approval is required **Requires EQ Biosolids		Soils Information Date of last soils analysis _____ *Phosphorous _____ units _____ Potassium _____ units _____ pH _____ Soil type(s) _____ *300 lb/ac Bray P1 maximum								
Expected crops/vegetation to be grown <table border="1" style="width:100%"> <tr> <th>Crop/Vegetation</th> <th>Nitrogen Requirement</th> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> </table>		Crop/Vegetation	Nitrogen Requirement	_____	_____ lbs. N/acre	_____	_____ lbs. N/acre	_____	_____ lbs. N/acre	Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☐ *Higher (Requires an approved site reclamation plan)
Crop/Vegetation	Nitrogen Requirement									
_____	_____ lbs. N/acre									
_____	_____ lbs. N/acre									
_____	_____ lbs. N/acre									
Total acreage of site _____ Acreage used for crops _____										
Public Exposure Is this land with a high potential for public exposure? (See Definitions section on Page 19) ☐ Yes ☐ No		If agricultural, is the site tilled? ☐ Yes ☐ No ☐ NA								

On an attached sheet provide a copy of the following:

- A Farm Services Agency map (formerly ASCS) of the proposed application site with property boundaries, surface waters, and discharge location of field tiles when present.
- A copy of the soil test analysis that is no older than 2 years old at the time of land application.
- A copy of the plat map (or other map showing property dimensions) with the proposed land application site highlighted.
- A soil survey map with the land application site identified.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****Land Application Site List**

Provide the following information for every new or existing site that you intend to use in the next five years (biosolids permit cycle). Each listed site should either have had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998, or the completed forms should be included with this document. To add new sites to the Land Application Site List during the biosolids permit cycle, modify your RMP by submitting a completed DEQ Site Identification Form with attachments, a copy of your public notification letter and wait the required ten-day notification period. Use additional sheets as necessary.

PLEASE TYPE OR PRINT

Facility Name Cadillac Waste Water Treatment Plant				NPDES or COC PERMIT NUMBER MI0020257		
Site Identification Number	Latitude	Longitude	Acres	Owners Last Name	New Site? (check if yes)	CPLR Site? (check if yes)
IV01	N44 14.567'	W85 17.589'	33	VANDERVLUGHT	☐	☐
IV02	N44 14.429'	W85 17.414'	18	VANDERVLUGHT	☐	☐
IV03	N44 14.308'	W85 17.589'	37	VANDERVLUGHT	☐	☐
IV04	N44 14.000'	W85 16.900'	16	VANDERVLUGHT	☐	☐
IV05	N44 13.940'	W85 16.900'	16	VANDERVLUGHT	☐	☐
IV06	N44 13.877'	W85 16.900'	16	VANDERVLUGHT	☐	☐
IV07	N44 13.811'	W85 16.900'	14	VANDERVLUGHT	☐	☐
IV08	N44 13.735'	W85 16.900'	15	VANDERVLUGHT	☐	☐
IV09	N44 14.030'	W85 16.550'	8	VANDERVLUGHT	☐	☐
IV10	N44 13.940'	W85 16.471'	7	VANDERVLUGHT	☐	☐
IV11	N44 14.429'	W85 17.360'	12	VANDERVLUGHT	☐	☐
IV12	N44 14.429'	W85 17.300'	6	VANDERVLUGHT	☐	☐
IV13	N44 14.429'	W85 17.240'	6	VANDERVLUGHT	☐	☐
IV14	N44 14.429'	W85 17.625'	8	VANDERVLUGHT	☐	☐
IV15	N14 14.279'	W85 17.035'	6	VANDERVLUGHT	☐	☐
KL01	N14 14.279'	W85 17.035'	46	LIEDEKE	☐	☐
CB01	N44 15.747'	W85 15.649'	25	BALDWIN	☐	☐
CB02	N44 15.696'	W85 15.579'	23	BALDWIN	☐	☐
					☐	☐
					☐	☐
					☐	☐

RESIDUALS MANAGEMENT PROGRAM**SECTION IV – SIGNATURE PAGE**

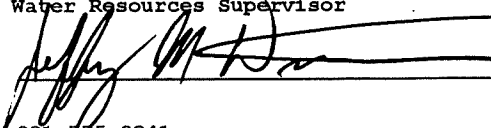
PLEASE TYPE OR PRINT

Facility Name Cadillac Waste Water Treatment Plant	NPDES or COC PERMIT NUMBER MI0020257
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1. CERTIFICATION

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with the system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Name and official title Jeffrey M. Dietlin
Water Resources Supervisor

Signature 

Telephone number 231-775-2841

Date signed 7/26/00

Upon request from the State, you may be required to submit additional information necessary to access biosolids use or disposal practices at your facility or to identify appropriate permitting requirements.

SEND COMPLETED FORMS TO THE APPROPRIATE DISTRICT OFFICE LISTED IN APPENDIX A.